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On the systematic underestimation of methane conversion factors in IPCC guidance

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ABSTRACT

This paper documents a systematic underestimation in how the Tier 2 methane conversion factors (MCF) are calculated in IPCC (2006) guidelines for liquid manure management. The first issue is the use of annual average temperature as an input to a non-linear function describing methane production. As expected based on Jensen's inequality, the MCF calculated based on annual average temperature is always an underestimate. In regions with large intra-annual temperature ranges, such as temperate climates, the underestimation can exceed 30%. A second issue is the lack of consideration for volatile solids retention time. Future updates to the IPCC methodology should therefore account for intra-annual temperature regime and retention time—not simply annual average temperature.

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1. Introduction

The Intergovernmental Panel on Climate Change (IPCC) provides guidance for calculating methane (CH_4) emissions from liquid manure management. In Chapter 10.4 of the guidance document (IPCC, 2006), the procedure for calculating CH_4 emissions from liquid manure management is described. The Tier 1 and Tier 2 procedures represent the most straight forward calculations, and are therefore most likely to be used by countries around the globe.

The Tier 1 method uses livestock population data and a temperature-based emission factor to estimate emissions, and the IPCC suggests it is *good practice* to use average annual temperature. The IPCC Tier 2 method is more detailed and is recommended for use when a specific livestock category is a substantial part of a nation's emissions. The Tier 2 approach requires livestock population data, temperature information, manure characteristics, and manure management system characteristics. The Tier 1 emission factor defines CH_4 emissions from manure management in terms of $\text{kg CH}_4 \text{ head}^{-1} \text{ year}^{-1}$, while the Tier 2 approach uses calculation of volatile solids excretion for each animal group combined with a methane conversion factor (MCF) which determines the conversion of volatile solids to CH_4 .

Therefore, both Tier 1 and Tier 2 methods are temperature dependent. The Tier 1 emission factors are provided based on annual average temperature in 1°C increments from $\leq 10^\circ\text{C}$ to $\geq 28^\circ\text{C}$ (Table 10.14 in IPCC, 2006). For example, North American

dairy cows are given an EF ranging from 48 to $112 \text{ kg CH}_4 \text{ head}^{-1} \text{ yr}^{-1}$, which is based on a weighted average for a range of manure management systems including lagoons, liquid, solid, etc. (Annex 10A.2 in IPCC, 2006). Therefore, the Tier 1 EFs are based on the Tier 2 calculations with assumptions about the manure management system distribution ("MS") for each livestock class and region. The Tier 2 method uses an equation to calculate an emission factor, where the key temperature-dependent parameter is MCF. The default MCF values for liquid manure ("Liquid/Slurry") are provided based on annual average temperature in 1°C increments from $\leq 10^\circ\text{C}$ to $\geq 28^\circ\text{C}$ (Table 10.17 in IPCC, 2006), resulting in a range from 17 to 80%. These values are based on IPCC expert judgement in combination with Mangino et al. (2001).

The report by Mangino et al. (2001) describes a van't Hoff-Arrhenius equation to characterize the temperature response of methane producing microorganisms in a factor denoted as " f ". The equation is calculated on a monthly basis, using monthly average air temperature. After calculating f in each month and determining CH_4 emissions each month, the annual sum of CH_4 emissions are calculated to determine the annual MCF.

Whereas Mangino et al. (2001) calculate the f parameter monthly and then calculate annual MCF based on the sum of monthly calculations, the (IPCC, 2006) approach calculates the annual MCF based on the annual average temperature. This short communication focuses on the simple but critical error of using annual average temperature when determining the annual MCF (Tier 2) and by extension the annual EF (Tier 1). This issue is an example of the mathematical principle called *Jensen's inequality* (Jensen, 1905) which applies to non-linear averaging, namely that

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the average output of concave-up functions with continuous input is always greater than the output of the function using the average input (e.g. Chesson, 1996; Azar, 2008). This paper will show that the error causes a systematic underestimation of the MCF and EF.

This paper sets aside other potential concerns, criticisms, or improvements that could be made to the IPCC approach to manure CH₄ emissions, and focuses only on the internal consistency of IPCC calculations. The issue of non-linear averaging is important because CH₄ production is inherently a non-linear response to temperature. Therefore, future improvements in the IPCC Guidelines must handle this subject properly.

2. IPCC temperature response

The MCF for manure management categories *Liquid/Slurry* and *Pit storage below animal confinements (>1 month)* from Table 10.17 in IPCC (2006) are shown in Table 1. In addition, the f parameter has been calculated using the van't Hoff-Arrhenius equation from Mangino et al. (2001):

$$f = \exp \left[\frac{E(T_2 - T_1)}{RT_1T_2} \right] \quad (1)$$

where T_1 is 303.16 K, T_2 is the ambient temperature (K), E is activation energy (15,175 cal mol⁻¹), and R is the ideal gas constant (1.987 cal K⁻¹ mol⁻¹).

Clearly the IPCC MCF values have been calculated using the annual average temperature and the equation from Mangino et al. (2001). All values in Table 1 are identical, except that the (IPCC, 2006) MCF for ≥ 28 °C is less than the value of f calculated at 28 °C.

3. Systematic underestimation

Now consider what the MCF should be when calculated using monthly temperature data, rather than the annual average. To this end, let us begin with the temperature regime shown in the (Mangino et al., 2001) paper (Table 2). In this case the temperature value and f values have been entered exactly as shown in the paper.

Table 1

Annual average temperature values provided in IPCC (2006) (°C), converted temperature (K), the f parameter calculated according to Eq. (1) for each temperature, and the MCF value given in IPCC (2006) (Table 10.17) for liquid manure. Temperatures denoted with an asterisk are reported in the IPCC table as ≤ 10 and ≥ 28 °C. See supplemental materials for a spreadsheet with calculations.

Annual average temperature		Calculated	IPCC (2006)
°C	K	f	MCF (%)
10*	283.15	0.17	17
11	284.15	0.19	19
12	285.15	0.20	20
13	286.15	0.22	22
14	287.15	0.25	25
15	288.15	0.27	27
16	289.15	0.30	29
17	290.15	0.32	32
18	291.15	0.35	35
19	292.15	0.39	39
20	293.15	0.42	42
21	294.15	0.46	46
22	295.15	0.50	50
23	296.15	0.55	55
24	297.15	0.60	60
25	298.15	0.65	65
26	299.15	0.71	71
27	300.15	0.78	78
28*	301.15	0.85	80

Table 2

Temperature and f parameter for twelve months as shown in Mangino et al. (2001) for Iowa. Temperatures for December, January, and February were set to 5 °C by Mangino et al. (2001) for modeling purposes. See supplemental materials for a spreadsheet with calculations.

Month	Monthly average air temperature		f
	K	°C	
October	283.3	10.1	0.17
November	279.8	6.6	0.12
December	278.2	5.0	0.10
January	278.2	5.0	0.10
February	278.2	5.0	0.10
March	279.1	5.9	0.11
April	282.5	9.4	0.16
May	290.0	16.8	0.32
June	292.8	19.6	0.41
July	295.4	22.2	0.51
August	295.5	22.4	0.52
September	290.9	17.7	0.34
Average	285.3	12.1	0.25

The annual average value of f calculated by averaging monthly values is 0.25 (Table 2). On the other hand, the annual average temperature used by Mangino et al. is 12.1 °C (Table 2), which gives a calculated f of 0.20 (see Table 1). Therefore, by using the annual average temperature instead of monthly temperature, the f parameter is underestimated by 20% (0.20 vs 0.25). This result is exactly as predicted by Jensen's inequality since the Arrhenius function is concave up.

It is worth noting that a minimum winter temperature of 5 °C was imposed by Mangino et al. (2001) for modeling purposes. Therefore the "annual average" temperature based on the Mangino et al. monthly data (12.1 °C) is greater than the actual annual average (e.g., 9.0 °C annual average for 1981–2010 at the Ames, Iowa municipal airport; www.ncdc.noaa.gov). If one uses the actual annual temperature (i.e., 9.0 °C) the result would be an even lower MCF of 0.15 using equation 1, or 0.17 using the (IPCC, 2006) lookup table (Table 1). Therefore, the underestimation would be even larger (i.e. 0.17 vs 0.25).

It is also interesting to note that the MCF calculated in the example of Mangino et al. (2001) was 0.70 (for an Anaerobic Lagoon), which is more than 4-fold greater than the default IPCC value for Liquid/Slurry with an annual average temperature ≤ 10 °C. This MCF was obtained using the same dataset and the same f values shown in Table 2, while also considering the amount of VS available for degradation each month. The annual MCF calculated by Mangino et al. reached 0.70 because of the long VS retention time in an anaerobic lagoon that was cleaned out once per year. With increasing retention time, MCF will be influenced by the quantity of accumulated volatile solids available in storage during the warm period, not simply the average f value during the year. Compared to Anaerobic Lagoons, Liquid/Slurry manure handling systems have different designs and typically shorter VS retention times (around 6 months), but retention time must still play a role in determining the MCF for Liquid/Slurry systems. This highlights another key oversight in the (IPCC, 2006) guidance for Liquid/Slurry systems, which is that VS retention time was not taken into account.

Next let us turn to a more general assessment of Jensen's inequality in the context of calculating the f parameter. The table below demonstrates this principle with several other temperature regimes with similar annual average temperatures (12.2 °C) but differing intra-annual variability. As expected, the annual f parameter (and in-turn the annual MCF) is systematically underestimated in all three temperature regimes. The extent of underestimation scales with the intra-annual variability, ranging

Table 3

Temperature and f parameter for twelve months under three hypothetical temperature regimes, with similar annual average temperature but differing levels of variation. SD is standard deviation, and amplitude is $\frac{1}{2}$ of the range. See [supplemental materials](#) for a spreadsheet with calculations.

Month	Large variation			Moderate variation			Low variation		
	Tav (K)	Tav (°C)	f	Tav (K)	Tav (°C)	f	Tav (K)	Tav (°C)	f
October	286	12.9	0.22	283.3	10.2	0.17	285	11.9	0.20
November	282	8.9	0.15	281	7.9	0.14	283	9.9	0.17
December	275	1.9	0.08	279	5.9	0.11	279	5.9	0.11
January	274	0.9	0.07	279	5.9	0.11	280	6.9	0.12
February	275	1.9	0.08	278.2	5.1	0.10	283	9.9	0.17
March	279.1	6.0	0.11	279.1	6.0	0.11	285	11.9	0.20
April	282.5	9.4	0.16	282.5	9.4	0.16	287	13.9	0.24
May	290	16.9	0.32	290	16.9	0.32	288	14.9	0.27
June	295	21.9	0.50	292.8	19.7	0.41	289	15.9	0.29
July	297	23.9	0.59	294	20.9	0.46	290	16.9	0.32
August	296	22.9	0.54	294	20.9	0.46	288	14.9	0.27
September	292	18.9	0.38	290.9	17.8	0.35	287	13.9	0.24
Annual average	285.3	12.2	0.27	285.3	12.2	0.24	285.3	12.2	0.22
SD	8.6	8.6		6.5	6.5		3.5	3.5	
Amplitude		11.5			7.9			5.5	
f from Annual Tav _g			0.21			0.21			0.21
Underestimation			23%			14%			4%

from 4% in the low-variation scenario to 23% in the high-variation scenario (see [Table 3](#)).

To further assess the underestimation in general terms, a synthetic temperature regime was created by selecting a mean annual temperature and superimposing a sinusoidal function to describe the annual temperature fluctuation over 12 months. Both the annual average and the amplitude of the sine function were changed to cover a range of possible conditions. For context, the temperature regimes (mean $\pm$ amplitude) in several locations are shown below in order of decreasing latitude (see [supplemental materials](#) for a spreadsheet with calculations and data from http://climate.weather.gc.ca/climate_normals; www.ncdc.noaa.gov; www.meteostat.net). As expected, the annual average temperature increases with decreasing latitude, while the amplitude decreases.

- Edmonton, ON, Canada, 1981–2010: 2.6 ± 14.2 °C
- Ottawa, ON, Canada, 1981–2010: 6.6 ± 15.7 °C
- Ames, IA, USA, 1981–2010: 9.0 ± 14.9 °C
- Chino, CA, USA, 1981–2010: 17.5 ± 6.9 °C
- Mexico City, MX, 2015: 17.3 ± 2.7 °C
- Villavicencio, Columbia, 2015: 26.3 ± 1.3 °C.

For each scenario, the f parameter was calculated based on monthly temperature and calculated again based on annual average temperature to determine the extent of underestimation

Table 4

The effect of annual mean temperature and intra-annual temperature range (amplitude of a sine function) on the percentage underestimation of the annual f parameter. Underestimation (%) was calculated by comparing the f parameter calculated based on annual average temperature to the annual average f parameter calculated based on monthly average temperature. See [supplemental materials](#) for a spreadsheet with calculations.

Annual mean temperature (°C)	Amplitude of intra-annual temperature range		
	± 5 °C	± 10 °C	± 15 °C
6	5%	19%	35%
10	5%	18%	34%
14	5%	17%	33%
18	5%	16%	31%
22	4%	15%	30%
26	4%	15%	29%

([Table 4](#)). From this assessment, it is clear that the critical parameter is the intra-annual temperature range. On the other hand, the mean annual temperature has only a small inverse relationship with the underestimation ([Table 4](#)).

4. Conclusion

This short communication documented a systematic underestimation in how the Tier 2 MCF values for Liquid/Slurry and Pit storages are calculated in the 2006 IPCC Guidelines for National Greenhouse Gas Inventories. This underestimation also affects Tier 1 emission factors for manure emissions, which are partially based on Tier 2 MCF values. The core issue is the use of annual average temperature as an input to a non-linear function describing CH₄ production. As expected based on Jensen's inequality, the MCF calculated based on annual average temperature is always an underestimate, and this paper has shown that the magnitude of the underestimation is significant, especially in regions with large intra-annual temperature ranges. A second issue is the lack of consideration for volatile solids retention time. Future updates to the IPCC methodology should therefore account for intra-annual temperature regime and retention time when determining Tier 1 and Tier 2 guidance.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.wasman.2018.01.037>.

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